The n-th homotopy group of the n-sphere is the integers

The Univalent Foundations Program

hott-book:homotopy.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author thm environment at source lines 1855-1857 and its proof environment opens on line 1859, the line after the blank separator; the pairing is by adjacency and the retained proof range 1859-1865 is the whole environment.

Editorial note. The pinned book label-number dump carries no entry for this label, so no counter value is imposed and the counted theorem prints with the wrapper numbering; the original number is not claimed.

Editorial note. Three labels the proof cites are declared in external_references as non-load-bearing navigation pointers, all in this chapter: cor:pi1s1 and cor:pis2-hopf supply the two base cases and cor:stability-spheres is the theorem applied in the inductive step. Each is an already established result of the book and none of them is the step carried out here.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the homotopy groups of the spheres on the stable diagonal (author label thm:pinsn).

Allowed standard prerequisites (editorial).

- The first two entries of the diagonal. The fundamental group of the circle is the integers, and the second homotopy group of the two-sphere is the integers.. They are the two base cases of the induction.
- Stability for spheres. For k at most 2n-2 the group in degree k+1 of the (n+1)-sphere equals the group in degree k of the n-sphere.. It is the inductive step, applied with k equal to n.
- Homotopy groups and truncations. The n-th homotopy group of a pointed type is the 0-truncation of its n-fold loop space.. It is the definition through which the two sides of the inductive step are compared.

Source reference. Original corollary 8.1.11: sources/hott-homotopy/pinned-original-source.tex, source lines 643–643. Book Corollary 8.1.11, the fundamental group of the circle, used as the first base case of the induction. Navigation pointer only.

Source reference. Original corollary 8.5.2: sources/hott-homotopy/pinned-original-source.tex, source lines 1217–1217. Book Corollary 8.5.2, the second homotopy group of the two-sphere, used as the second base case. Navigation pointer only.

Source reference. Original corollary 8.6.15: sources/hott-homotopy/pinned-original-source.tex, source lines 1817–1817. Book Corollary 8.6.15, stability for spheres, applied in the inductive step with k equal to n. Navigation pointer only: the retained proof carries out the range check and the induction itself.

Theorem 0.0.1. $\pi_n(\mathbb{S}^n) = \mathbb{Z}$ *for every* $n \geq 1$.

Proof. The proof is by induction on n. We already have $\pi_1(\mathbb{S}^1) = \mathbb{Z}$ (original corollary 8.1.11) and $\pi_2(\mathbb{S}^2) = \mathbb{Z}$ (original corollary 8.5.2). When $n \geq 2$, $n \leq (2n - 2)$. Therefore, by original corollary 8.6.15, $\pi_{n+1}(S^{n+1}) = \pi_n(S^n)$, and this equivalence, combined with the inductive hypothesis, gives the result. □